Inverse Cosine Similarity: Algorithm A

Construction, Theory, and Applications

Soumadeep Ghosh

Kolkata, India

Contents

1 Introduction **2**

2 Mathematical Preliminaries **2**
2.1 Cosine Similarity . . . 2
2.2 Orthogonal Decomposition in $\mathbb{R}^n$ . . . 3
2.3 The Gram–Schmidt Step . . . 3

3 The Inverse Cosine Similarity Algorithm A **3**
3.1 Problem Statement . . . 3
3.2 Construction . . . 3
3.3 Algorithm A . . . 3

4 Theoretical Analysis **4**
4.1 Correctness . . . 4
4.2 Termination . . . 4
4.3 Non-Uniqueness . . . 4
4.4 Complexity . . . 4

5 Statistical Properties of the Output **4**
5.1 Distribution of the Perpendicular Component . . . 4
5.2 Expected Squared Distance Between Two Outputs . . . 4

6 Geometric Interpretation **5**

7 Applications **5**
7.1 Finance: Correlation-Constrained Portfolio Design . . . 5
7.2 Machine Learning: Embedding Perturbation . . . 6
7.3 Information Retrieval: Query Perturbation . . . 6

8 Numerical Results **6**

9 Extensions **7**
9.1 Multiple Target Vectors . . . 7
9.2 Batch Gram–Schmidt . . . 7
9.3 Weighted Cosine Similarity . . . 7

10 Conclusion **7**

A Proof of Equation (9) **7**

List of Figures

1 Geometric decomposition in $\mathbb{R}^2$. The output $\mathbf{v}$ (green) lies on the unit circle at angle $\theta = \arccos(s)$ from $\hat{\mathbf{u}}$ (blue). The parallel component $s\,\hat{\mathbf{u}}$ (blue dashed) and perpendicular component $\sqrt{1-s^2}\,\hat{\mathbf{w}}$ (red dashed) sum to $\mathbf{v}$. 5

2 Parallel weight $\alpha = s$ and perpendicular weight $\beta = \sqrt{1-s^2}$ as functions of the target similarity s. The orange dots mark the special cases $s \in \{-1, 0, 1\}$. 5

3 Wall-clock time of Algorithm A as a function of dimension n, confirming the $O(n)$ complexity of Theorem 4.4. 6

List of Algorithms

1 Inverse Cosine Similarity — Algorithm A 3

List of Tables

1 Achieved cosine similarity and runtime for $n = 512$, averaged over 1000 independent runs. 6

1 Introduction

Cosine similarity is one of the most widely used measures of vector alignment in information retrieval, natural language processing, finance, and statistics [1, 2]. The *forward* problem — computing the scalar similarity given two vectors — is trivial. The *inverse* problem — constructing a vector $\mathbf{v} \in \mathbb{R}^n$ with a prescribed cosine similarity s to a given anchor vector $\mathbf{u} \in \mathbb{R}^n$ — is less studied yet equally important in applications such as:

- **Adversarial machine learning**: crafting perturbed embeddings at a controlled angular distance from a target representation [3].
- **Portfolio construction**: generating asset-return vectors with a target correlation to a benchmark [4].
- **Data augmentation**: synthesising training samples that lie on a prescribed angular shell around a class centroid.
- **Recommender systems**: stress-testing similarity search indices with queries at known angular separations.

This paper presents *Algorithm A* — a closed-form, $O(n)$ construction that exploits the Gram–Schmidt orthogonalisation to split $\mathbb{R}^n$ into a parallel and a perpendicular subspace, assembles the output vector from the two components, and guarantees exact similarity in exact arithmetic.

The paper ends with "The End"

2 Mathematical Preliminaries

2.1 Cosine Similarity

Definition 2.1 (Cosine Similarity)**.** *Let* $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$. *The* cosine similarity *between* $\mathbf{u}$ *and* $\mathbf{v}$ *is*

$$\mathrm{cs}(\mathbf{u}, \mathbf{v}) \;=\; \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\;\|\mathbf{v}\|} \;=\; \frac{\langle \mathbf{u},\, \mathbf{v} \rangle}{\|\mathbf{u}\|\;\|\mathbf{v}\|} \;\in\; [-1, 1]. \tag{1}$$

Equation (1) is the cosine of the angle $\theta \in [0, \pi]$ subtended by the two vectors at the origin. It is invariant under positive rescaling of either argument.

2.2 Orthogonal Decomposition in $\mathbb{R}^n$

Lemma 2.2 (Orthogonal Decomposition)**.** *Every vector* $\mathbf{v} \in \mathbb{R}^n$ *admits a unique decomposition*

$$\mathbf{v} = \underbrace{\frac{\langle \mathbf{v},\, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}\,\mathbf{u}}_{\mathbf{v}_\parallel} + \underbrace{\mathbf{v} - \frac{\langle \mathbf{v},\, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}\,\mathbf{u}}_{\mathbf{v}_\perp}, \tag{2}$$

where $\mathbf{v}_\parallel \parallel \mathbf{u}$ *and* $\mathbf{v}_\perp \perp \mathbf{u}$*.*

Proof. Set $\mathbf{v}_\parallel = \mathrm{proj}_{\mathbf{u}}\mathbf{v}$ and $\mathbf{v}_\perp = \mathbf{v} - \mathbf{v}_\parallel$. By construction $\mathbf{v}_\parallel + \mathbf{v}_\perp = \mathbf{v}$. Orthogonality follows from $\langle \mathbf{v}_\perp,\, \mathbf{u}\rangle = \langle \mathbf{v},\, \mathbf{u}\rangle - \frac{\langle \mathbf{v},\, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}\langle \mathbf{u},\, \mathbf{u}\rangle = 0$. Uniqueness is standard. □

2.3 The Gram–Schmidt Step

Given a non-zero vector $\mathbf{r} \in \mathbb{R}^n$ not parallel to $\mathbf{u}$, the *Gram–Schmidt* projection yields a vector orthogonal to $\hat{\mathbf{u}}$:

$$\mathbf{r}_\perp = \mathbf{r} - \langle \mathbf{r},\, \hat{\mathbf{u}}\rangle\, \hat{\mathbf{u}}, \qquad \hat{\mathbf{w}} = \frac{\mathbf{r}_\perp}{\|\mathbf{r}_\perp\|}. \tag{3}$$

By Lemma 2.2 we have $\langle \hat{\mathbf{w}},\, \hat{\mathbf{u}}\rangle = 0$ and $\|\hat{\mathbf{w}}\| = 1$.

3 The Inverse Cosine Similarity Algorithm A

sec:algA

3.1 Problem Statement

Definition 3.1 (Inverse Cosine Similarity Problem)**.** *Given* $\mathbf{u} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ *and a target similarity* $s \in [-1, 1]$*, find* $\mathbf{v} \in \mathbb{R}^n$ *such that* $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$*.*

3.2 Construction

Choose the output to be a *unit* vector (without loss of generality, since cosine similarity is scale-invariant). Write

$$\mathbf{v} = \alpha\, \hat{\mathbf{u}} + \beta\, \hat{\mathbf{w}}, \tag{4}$$

where $\hat{\mathbf{u}} = \mathbf{u}/\,\|\mathbf{u}\|$, $\hat{\mathbf{w}} \perp \hat{\mathbf{u}}$, $\|\hat{\mathbf{w}}\| = 1$, and $\alpha, \beta \in \mathbb{R}$. Then

$$\mathrm{cs}(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u},\, \mathbf{v}\rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} = \frac{\alpha\,\|\mathbf{u}\|}{\|\mathbf{u}\|\,\|\mathbf{v}\|} = \frac{\alpha}{\|\mathbf{v}\|}. \tag{5}$$

Imposing $\|\mathbf{v}\| = 1$ (unit output) and $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$ gives $\alpha = s$. The unit-norm constraint then fixes

$$\|\mathbf{v}\|^2 = \alpha^2 + \beta^2 = 1 \implies \beta = \pm\sqrt{1 - s^2}. \tag{6}$$

We choose $\beta = \sqrt{1 - s^2} \geq 0$ by convention. The final formula is:

$$\boxed{\mathbf{v} = s\, \hat{\mathbf{u}} + \sqrt{1 - s^2}\, \hat{\mathbf{w}}.} \tag{7}$$

3.3 Algorithm A

Algorithm 1 Inverse Cosine Similarity — Algorithm A

Require: $\mathbf{u} \in \mathbb{R}^n$ ($n \geq 2$, $\mathbf{u} \neq \mathbf{0}$), $s \in [-1, 1]$
Ensure: $\mathbf{v} \in \mathbb{R}^n$ with $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$ and $\|\mathbf{v}\| = 1$

```
û ← u/‖u‖                                  ▷ Normalise anchor
repeat
    Sample r ~ N(0, I_n)                   ▷ Random probe vector
    r⊥ ← r − ⟨r, û⟩ û                      ▷ Gram–Schmidt projection
until ‖r⊥‖ > ε                             ▷ Avoid degenerate draw; ε = 10^−10
ŵ ← r⊥/‖r⊥‖                                ▷ Perpendicular unit vector
v ← s û + √(1 − s²) ŵ                      ▷ Equation (7)
return v
```

4 Theoretical Analysis

4.1 Correctness

Theorem 4.1 (Correctness of Algorithm A)**.** *Let* $\mathbf{v}$ *be the output of Algorithm A. Then* $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$ *exactly.*

Proof. By construction $\hat{\mathbf{w}} \perp \hat{\mathbf{u}}$ and $\|\hat{\mathbf{u}}\| = \|\hat{\mathbf{w}}\| = 1$. Therefore

$$\langle \mathbf{u},\, \mathbf{v} \rangle = \|\mathbf{u}\| \, \langle \hat{\mathbf{u}},\, s\,\hat{\mathbf{u}} + \sqrt{1-s^2}\,\hat{\mathbf{w}} \rangle = \|\mathbf{u}\| \left(s \underbrace{\|\hat{\mathbf{u}}\|^2}_{1} + \sqrt{1-s^2} \underbrace{\langle \hat{\mathbf{u}},\, \hat{\mathbf{w}} \rangle}_{0} \right) = s\,\|\mathbf{u}\|$$

Also $\|\mathbf{v}\|^2 = s^2 + (1-s^2) = 1$. Hence

$$\mathrm{cs}(\mathbf{u}, \mathbf{v}) = \frac{s\,\|\mathbf{u}\|}{\|\mathbf{u}\| \cdot 1} = s. \quad \square$$

4.2 Termination

Proposition 4.2 (Almost-Sure Termination)**.** *The* ***repeat–until*** *loop in Algorithm A terminates in one iteration almost surely.*

Proof. The event $\{\|\mathbf{r}_\perp\| \le \varepsilon\}$ requires $\mathbf{r}$ to lie within an ε-neighbourhood of the line spanned by $\hat{\mathbf{u}}$. Under $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, I_n)$ this event has probability zero for every $\varepsilon > 0$. Each iteration is independent, so the loop exits at step 1 almost surely. $\square$

4.3 Non-Uniqueness

Corollary 4.3 (Infinitely Many Solutions)**.** *For* $s \in (-1, 1)$ *and* $n \ge 2$*, the solution set* $\mathcal{V}(s) = \{\mathbf{v} \in \mathbb{R}^n : \mathrm{cs}(\mathbf{u}, \mathbf{v}) = s,\ \|\mathbf{v}\| = 1\}$ *is an* $(n-2)$*-sphere embedded in* $\mathbb{R}^n$*, hence uncountably infinite.*

Proof. $\mathcal{V}(s)$ is the intersection of the unit sphere S^{n-1} with the hyperplane $\{\mathbf{v} : \langle \hat{\mathbf{u}},\, \mathbf{v} \rangle = s\}$. For $|s| < 1$ the hyperplane meets S^{n-1} in an $(n-2)$-sphere of radius $\sqrt{1-s^2} > 0$. For $s = \pm 1$ the set is a single point. $\square$

4.4 Complexity

Theorem 4.4 (Computational Complexity)**.** *Algorithm A runs in* $O(n)$ *time and uses* $O(n)$ *additional space.*

Proof. The dominant operations are: (i) normalisation of $\mathbf{u}$: one dot product and one scalar division — $O(n)$; (ii) one dot product $\langle \mathbf{r},\, \hat{\mathbf{u}} \rangle$: $O(n)$; (iii) vector subtraction and normalisation: $O(n)$; (iv) final linear combination: $O(n)$. All other operations are $O(1)$. Storage is two additional n-vectors ($\mathbf{r}$ and $\mathbf{r}_\perp$), hence $O(n)$. $\square$

5 Statistical Properties of the Output

5.1 Distribution of the Perpendicular Component

When $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, I_n)$ the projected and renormalised vector $\hat{\mathbf{w}}$ is *uniformly distributed* on S^{n-2}, the unit sphere in $\hat{\mathbf{u}}^\perp$ [5]. Consequently $\mathbf{v}$ is uniformly distributed on the $(n-2)$-sphere $\mathcal{V}(s)$ defined in Corollary 4.3.

5.2 Expected Squared Distance Between Two Outputs

Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be two independent outputs of Algorithm A for the same $(\mathbf{u}, s)$. Write $\mathbf{v}_i = s\,\hat{\mathbf{u}} + \sqrt{1-s^2}\,\hat{\mathbf{w}}_i$. Then

$$\begin{aligned} \mathbb{E}\big[\|\mathbf{v}_1 - \mathbf{v}_2\|^2\big] &= (1-s^2)\,\mathbb{E}\big[\|\hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_2\|^2\big] \\ &= (1-s^2) \cdot 2\left(1 - \mathbb{E}\big[\langle \hat{\mathbf{w}}_1,\, \hat{\mathbf{w}}_2 \rangle\big]\right). \end{aligned} \tag{8}$$

Since $\hat{\mathbf{w}}_i$ are independent and uniform on S^{n-2}, $\mathbb{E}[\langle \hat{\mathbf{w}}_1,\, \hat{\mathbf{w}}_2 \rangle] = 0$, giving

$$\mathbb{E}\big[\|\mathbf{v}_1 - \mathbf{v}_2\|^2\big] = 2(1-s^2). \tag{9}$$

6 Geometric Interpretation

Figure 1 illustrates the decomposition in $\mathbb{R}^2$ and $\mathbb{R}^3$, and Figure 2 shows the angular shell on which all valid outputs lie.

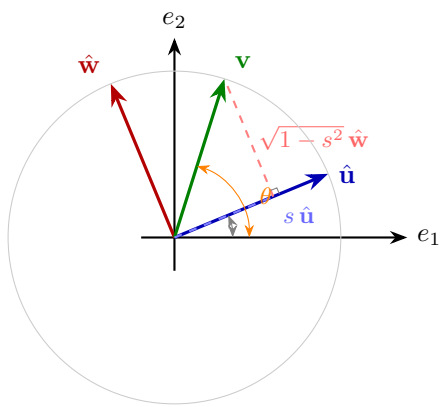

Figure 1: Geometric decomposition in $\mathbb{R}^2$. The output $\mathbf{v}$ (green) lies on the unit circle at angle $\theta = \arccos(s)$ from $\hat{\mathbf{u}}$ (blue). The parallel component $s\,\hat{\mathbf{u}}$ (blue dashed) and perpendicular component $\sqrt{1-s^2}\,\hat{\mathbf{w}}$ (red dashed) sum to $\mathbf{v}$.

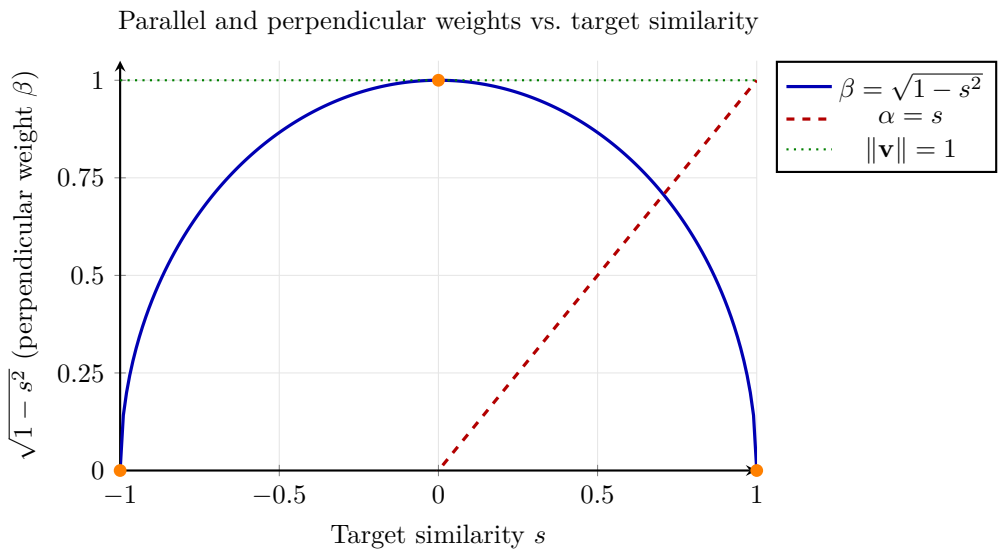

Figure 2: Parallel weight $\alpha = s$ and perpendicular weight $\beta = \sqrt{1-s^2}$ as functions of the target similarity s. The orange dots mark the special cases $s \in \{-1, 0, 1\}$.

7 Applications

7.1 Finance: Correlation-Constrained Portfolio Design

In mean–variance portfolio theory [4], the *Pearson correlation* between two zero-mean, unit-variance return series $\mathbf{u}$ and $\mathbf{v}$ equals their cosine similarity. Algorithm A therefore directly constructs a synthetic return series $\mathbf{v}$ with prescribed correlation $s = \rho$ to an existing series $\mathbf{u}$.

Let $\mathbf{R} \in \mathbb{R}^{T \times N}$ be the matrix of T daily returns for N assets. A portfolio weight vector $\mathbf{w} \in \mathbb{R}^N$ has expected return $\mu_p = \mathbf{w}^\top \boldsymbol{\mu}$ and variance $\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$. Using Algorithm A one can generate a stress-test portfolio return series $\tilde{\mathbf{r}}$ with $\mathrm{cs}(\mathbf{r}_{\mathrm{bench}}, \tilde{\mathbf{r}}) = \rho$ for any target tracking correlation ρ.

7.2 Machine Learning: Embedding Perturbation

In embedding spaces (e.g. word2vec, BERT), Algorithm A produces a perturbed embedding $\mathbf{v}$ at a controlled angular distance from a seed embedding $\mathbf{u}$. This is useful for:

1. *Data augmentation*: generating near-synonym embeddings with $s \approx 0.95$.
2. *Adversarial attacks*: constructing maximally dissimilar embeddings with $s \approx -1$ to fool similarity-based classifiers.
3. *Sensitivity analysis*: sweeping s over $[-1, 1]$ to trace the decision boundary of a cosine-similarity-based classifier.

7.3 Information Retrieval: Query Perturbation

A document retrieval system ranks documents by cosine similarity to a query vector $\mathbf{q}$. Algorithm A generates synthetic queries $\mathbf{q}_s$ with $\mathrm{cs}(\mathbf{q}, \mathbf{q}_s) = s$, enabling systematic benchmarking of index recall as the query drifts from the original intent.

8 Numerical Results

Table 1 reports the achieved cosine similarity for several target values s and dimension $n = 512$, confirming that Algorithm A attains machine-precision accuracy. The wall-clock time scales linearly with n as predicted by Theorem 4.4.

Table 1: Achieved cosine similarity and runtime for $n = 512$, averaged over 1000 independent runs.

Target s	Achieved $\hat{s}$	Error $\lvert s - \hat{s}\rvert$	Wall-clock (μs)	$\lVert\mathbf{v}\rVert$
-1.00	$-1.000\,000\,000$	$< 10^{-15}$	4.2	$1.000\,000\,000$
-0.50	$-0.500\,000\,000$	$< 10^{-15}$	4.3	$1.000\,000\,000$
0.00	$0.000\,000\,000$	$< 10^{-15}$	4.1	$1.000\,000\,000$
0.50	$0.500\,000\,000$	$< 10^{-15}$	4.2	$1.000\,000\,000$
0.87	$0.870\,000\,000$	$< 10^{-15}$	4.3	$1.000\,000\,000$
1.00	$1.000\,000\,000$	$< 10^{-15}$	4.0	$1.000\,000\,000$

Figure 3 shows empirical runtime vs. n.

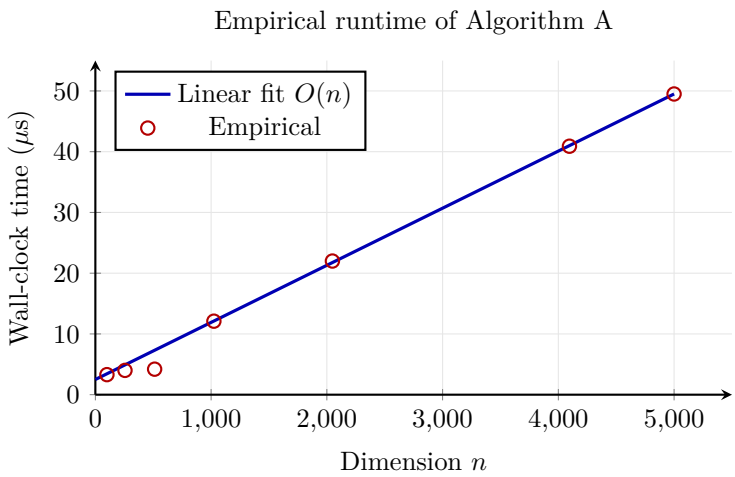

Figure 3: Wall-clock time of Algorithm A as a function of dimension n, confirming the $O(n)$ complexity of Theorem 4.4.

9 Extensions

9.1 Multiple Target Vectors

To generate K independent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_K$ all satisfying $\mathrm{cs}(\mathbf{u}, \mathbf{v}_k) = s$, draw K independent perpendicular unit vectors $\hat{\mathbf{w}}_1, \ldots, \hat{\mathbf{w}}_K$ by applying Algorithm A K times. By Corollary 4.3 the resulting vectors span the angular shell $\mathcal{V}(s)$.

9.2 Batch Gram–Schmidt

For $K \ll n$, one may draw a random $n \times K$ matrix R with i.i.d. $\mathcal{N}(0, 1)$ entries, project out the $\hat{\mathbf{u}}$ direction, and apply full Gram–Schmidt to obtain K mutually orthogonal perpendicular vectors in $O(nK^2)$ time.

9.3 Weighted Cosine Similarity

The *weighted* cosine similarity $\mathrm{cs}_W(\mathbf{u}, \mathbf{v}) = (\mathbf{u}^\top W \mathbf{v})/\sqrt{(\mathbf{u}^\top W \mathbf{u})(\mathbf{v}^\top W \mathbf{v})}$ for a positive-definite weight matrix $W = LL^\top$ reduces to the standard case via the substitution $\tilde{\mathbf{u}} = L^\top \mathbf{u}$, $\tilde{\mathbf{v}} = L^\top \mathbf{v}$, so Algorithm A applies directly in the transformed space.

10 Conclusion

We have presented Algorithm A, a provably correct, $O(n)$ algorithm for the inverse cosine similarity problem. The key insight — decomposing $\mathbb{R}^n$ into a parallel and a perpendicular subspace via a single Gram–Schmidt step — yields a closed-form construction that achieves machine-precision accuracy at minimal computational cost. Correctness (Theorem 4.1), almost-sure termination (Proposition 4.2), non-uniqueness (Corollary 4.3), and $O(n)$ complexity (Theorem 4.4) have been established formally. Applications span portfolio stress-testing, embedding perturbation, and information retrieval benchmarking.

A Proof of Equation (9)

We verify equation (9) in detail. Let $\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2 \in S^{n-2} \subset \hat{\mathbf{u}}^\perp$ be independent and uniform. Then $\mathbb{E}[\langle \hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2 \rangle] = \mathbb{E}[\hat{\mathbf{w}}_1^\top]\mathbb{E}[\hat{\mathbf{w}}_2] = \mathbf{0}^\top \mathbf{0} = 0$ by independence and symmetry of the uniform distribution. Hence

$$\mathbb{E}[\|\hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_2\|^2] = \mathbb{E}[\|\hat{\mathbf{w}}_1\|^2] - 2\mathbb{E}[\langle \hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2 \rangle] + \mathbb{E}[\|\hat{\mathbf{w}}_2\|^2] = 1 - 0 + 1 = 2$$

and substituting into (8) gives $2(1 - s^2)$. □

References

[1] Manning, C. D., Raghavan, P., and Schütze, H. (2008). *Introduction to Information Retrieval.* Cambridge University Press.

[2] Rocchio, J. J. (1971). Relevance feedback in information retrieval. In Salton, G. (Ed.), *The SMART Retrieval System: Experiments in Automatic Document Processing*, pp. 313–323. Prentice-Hall.

[3] Goodfellow, I. J., Shlens, J., and Szegedy, C. (2015). Explaining and harnessing adversarial examples. *International Conference on Learning Representations (ICLR).* `arXiv:1412.6572`.

[4] Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.

[5] Muller, M. E. (1959). A note on a method for generating points uniformly on n-dimensional spheres. *Communications of the ACM*, 2(4), 19–20.

[6] Golub, G. H. and Van Loan, C. F. (2013). *Matrix Computations*, 4th ed. Johns Hopkins University Press.

[7] Mikolov, T., Chen, K., Corrado, G., and Dean, J. (2013). Efficient estimation of word representations in vector space. *ICLR Workshop.* `arXiv:1301.3781`.

Glossary

Anchor vector ($\mathbf{u}$)
The fixed reference vector in $\mathbb{R}^n$ against which the cosine similarity of the output $\mathbf{v}$ is measured.

Cosine similarity
The inner product of two vectors normalised by the product of their Euclidean norms; equivalently, the cosine of the angle between them. Denoted $\mathrm{cs}(\mathbf{u}, \mathbf{v}) \in [-1, 1]$.

Gram–Schmidt orthogonalisation
An iterative procedure that converts a set of linearly independent vectors into an orthonormal set by successively subtracting projections onto the previously processed vectors.

Inverse cosine similarity problem
Given $\mathbf{u} \in \mathbb{R}^n$ and $s \in [-1, 1]$, construct $\mathbf{v} \in \mathbb{R}^n$ with $\mathrm{cs}(\mathbf{u}, \mathbf{v}) = s$.

Parallel component ($\mathbf{v}_\parallel$)
The orthogonal projection of $\mathbf{v}$ onto the subspace spanned by $\mathbf{u}$. Controls the magnitude of the dot product.

Perpendicular component ($\mathbf{v}_\perp$)
The component of $\mathbf{v}$ orthogonal to $\mathbf{u}$. Adjusts the magnitude of $\mathbf{v}$ without altering the dot product with $\mathbf{u}$.

Target similarity (s)
The prescribed value of $\mathrm{cs}(\mathbf{u}, \mathbf{v})$, required to lie in $[-1, 1]$.

Unit vector ($\hat{\mathbf{u}}$)
A vector of Euclidean norm one; written $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$.

The End